

■ Generative AI for Kids

A Fun Introduction to the AI That Creates!

Grade Level	Ages 8–14 (Grades 3–8)
Duration	60–90 Minutes
Subject	Technology · Computer Science · Media Literacy
Materials	Worksheet, coloured pens, internet-connected device (optional)

■ Learning Objectives

By the end of this lesson, students will be able to:

- Define *Artificial Intelligence* and *Generative AI* in their own words
- List at least three things Generative AI can create
- Give examples of Generative AI tools they may already use
- Explain how AI learns from data
- Discuss responsible and safe use of AI

SECTION 1

What Is Artificial Intelligence?

Artificial Intelligence — or **AI** for short — is when a computer is taught to do things that usually need human thinking. Reading, listening, drawing, writing, and even playing games are all things AI has learned to do!

Think of it this way...

When you were little, you learned to recognise cats by seeing many, many cats. Your brain stored all those examples. Now, when you see a new cat — even one you've never seen before — you know it's a cat! AI works the same way: it is shown millions of examples so it can learn patterns.

Types of AI (Simple Overview)

Type	What It Does	Example
Recognition AI	Identifies things in photos or sounds	Face unlock on a phone
Recommendation AI	Suggests things you might like	YouTube 'Up Next' videos
Generative AI	Creates brand-new text, images, music, or code	ChatGPT, DALL·E, Suno

➡ ■ Quick Activity 1 — Brain Warm-Up

Circle the things you think AI can do:

Write a poem · Feel sad · Compose music · Dream · Translate languages

Paint a picture · Fall in love · Answer questions · Write computer code

(Discuss your answers with a partner. There are no wrong answers yet!)

SECTION 2

What Is Generative AI?

Generative AI is a special kind of AI that doesn't just recognise or sort things — it actually *creates* brand-new content. You give it a prompt (a question or instruction) and it generates something original that never existed before.

■ **Prompt Example:** "Write a short story about a robot who is afraid of the dark."

The AI reads your words, thinks about everything it has learned, and then writes a completely new story — just for you!

What Can Generative AI Make?

■ **Text** — stories, poems, essays, emails

■ **Code** — computer programs

■■ **Images** — illustrations, paintings, photos

■ **Video** — short clips and animations

■ **Music** — songs, beats, jingles

■■ **Voice** — realistic-sounding speech

How Does It Learn?

Generative AI models are trained on **huge amounts of data** — billions of books, websites, images, and more. During training the model figures out patterns: which words usually follow other words, what makes a face look realistic, how melodies are structured. This training can take weeks and enormous amounts of computer power.

1	Collect Data	Gather text, images, or audio from the internet and books
2	Train the Model	The AI studies patterns across billions of examples
3	Fine-Tune	Humans help correct mistakes and teach better behaviour
4	You Use It!	Type a prompt → the AI generates a response

SECTION 3

Famous Generative AI Tools

Tool	Company	Creates	What It Does
ChatGPT	OpenAI	Text	Answers questions, writes stories and code, explains ideas
DALL·E / Midjourney	OpenAI / Midjourney	Images	Creates detailed pictures from a text description
Suno / Udio	Suno / Udio	Music	Composes full songs with lyrics from a short prompt
Gemini	Google	Text & Images	Google's AI assistant built into many Google products
Copilot	Microsoft / OpenAI	Code & Text	Helps programmers write code faster
ElevenLabs	ElevenLabs	Voice	Converts text into realistic human-sounding speech

■ Activity 2 — Prompt Engineer for a Day!

Imagine you are an AI. A friend gives you this prompt:

"Draw a friendly dragon reading a book in a cosy library."

In the box below, sketch or describe what your AI-generated picture would look like.

Then swap with a partner and compare!

SECTION 4

Using AI Responsibly

Generative AI is an amazing tool, but — like all tools — it needs to be used carefully. A hammer can build a house or break a window. AI can help you learn and create, or it can be misused. Here's how to be a responsible AI user:

■ DO	■ DON'T
■ Use AI to help you learn, not to cheat	■ Don't copy AI work and pretend it's yours
■ Always tell people when content was AI-made	■ Don't share fake AI images to fool people
■ Check AI facts — it can make mistakes!	■ Don't believe everything AI says without checking
■ Be kind: don't use AI to bully or hurt others	■ Don't use AI to create harmful or hateful content
■ Protect your privacy — don't share personal info	■ Don't give AI your name, address, or passwords

Can AI Ever Be Wrong?

Yes! AI can produce **hallucinations** — confident-sounding answers that are completely made up. For example, an AI might invent a fake book title and say a famous author wrote it. Always verify important facts using trusted sources like encyclopaedias, textbooks, or reliable news websites.

■ **Key Skill — Critical Thinking:** Treat AI output the way you treat a classmate's answer. It might be right, it might be wrong, it might be a bit of both. Your job is to check!

SECTION 5

The Future of Generative AI

Generative AI is growing incredibly fast. Researchers are exploring ways to use it in medicine, education, climate science, art, and much more. The students in this class will grow up in a world where AI is everywhere — understanding it now gives you a huge head start!

Exciting Future Uses

■ Medicine	AI helping doctors spot diseases earlier in scans and test results
■ Climate	AI designing cleaner energy systems and reducing waste
■ Education	AI tutors that adapt to exactly how you learn best
■ Space	AI analysing telescope data to find new planets
■ Accessibility	AI tools that help people with disabilities communicate and create

■ Activity 3 — Invent an AI!

Imagine you could build any Generative AI tool. What would it create?

My AI tool is called: _____

It creates:

It would help people by:

One rule people must follow when using it: _____

Share your idea with the class!

Key Words to Know

Word	What It Means
Artificial Intelligence (AI)	A computer system that can do tasks that normally need human intelligence.
Generative AI	AI that creates brand-new content such as text, images, music, or video.
Prompt	The instruction or question you give to an AI to tell it what to do.
Training Data	The huge collection of text, images, or audio the AI learns from.
Model	The AI system itself — the 'brain' that has learned from data.
Hallucination	When AI produces confident but incorrect or made-up information.
Algorithm	A set of step-by-step instructions that a computer follows.
Deep Learning	A type of AI training inspired by how the human brain works.
Output	The content the AI produces in response to your prompt.
Bias	Unfair patterns in AI results caused by unbalanced training data.

ASSESSMENT

Knowledge Check Quiz

Answer the following questions to show what you have learned. Use full sentences where asked.

1. What does the word 'generative' mean when we talk about Generative AI?

- a) It recognises pictures b) It creates brand-new content c) It searches the internet d) It repairs computers

2. Which of these is an example of a Generative AI output?

- a) A selfie you took b) An email a friend wrote c) A poem written by ChatGPT d) A book from the library

3. AI learns by studying large amounts of _____?

- a) Electricity b) Data c) Paper d) Paint

4. What is an AI 'hallucination'?

- a) A dream the AI has b) A picture of a ghost c) A confident but wrong answer d) A type of training method

5. Give one example of a responsible way to use Generative AI:

(Write your answer) _____

6. Why is it important to check facts that AI gives you?

(Write your answer) _____

7. In your own words, describe how Generative AI learns to create things:

(Write your answer — 2–3 sentences) _____

Answer Key (Multiple Choice): 1-b · 2-c · 3-b · 4-c

Teacher Notes & Extension Ideas

Suggested Lesson Plan (90 min)

Time	Phase	Activity
0–10 min	Warm-up	Activity 1 — Brain Warm-Up; class discussion on what AI can/can't do
10–25 min	Direct Instruction	Sections 1 & 2: What is AI? What is Generative AI? (slides + demo)
25–40 min	Exploration	Section 3: Explore real AI tools together as a class
40–55 min	Activity	Activity 2 — Prompt Engineer for a Day! (pairs)
55–70 min	Ethics Discussion	Section 4: Responsible use, hallucinations, fact-checking
70–80 min	Creative Task	Activity 3 — Invent an AI! (individual, then share)
80–90 min	Assessment	Quiz + class debrief; collect worksheets

Extension Activities

■ **Beginner:** Have students use a free AI chatbot to write a funny poem about their pet or favourite food, then read it aloud.

■■ **Intermediate:** Students generate an AI image of a fictional place, then write a paragraph describing it as if it were real.

■■■ **Advanced:** Students research one ethical concern about AI (job displacement, bias, deepfakes) and present a 2-minute argument for or against a related policy.

■ **Cross-Curricular:** English: Compare an AI-written poem with a human poem. Art: Study the aesthetics of AI-generated art. Science: Discuss AI in medical diagnosis.

■ Recommended Resources:

- *AI4K12 Initiative* — ai4k12.org (free curriculum)
- *Day of AI* — dayofai.org (MIT-created lessons)
- *Google's Teachable Machine* — teachablemachine.withgoogle.com
- *Common Sense Media AI resources* — commonsense.org/education